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> bootcamp.learn()
```

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JomCode Coding Bootcamp

CHAPTER 13

Relationships, joins, constraints and transactions

`learn / type / run / explain`

Plain meaning and technical explanation

PLAIN MEANING

Related tables connect records and rules prevent inconsistent data.

TECHNICAL EXPLANATION

Foreign keys reference another table. JOIN combines related rows. CHECK enforces rules. Transactions commit or roll back a group of changes.

REAL-WORLD USE

Understand posts, authors, post_chunks and safe database writes.

QUICK TIP

A transaction cannot roll back a separate storage upload; that needs cleanup code.

INSTALL / RUN

Run the full workbook in Supabase SQL Editor in a practice project. All tables here are temporary.

Worked example 1: Related rows

workbook.sql

```
insert into lesson_authors values (1, 'Ada'), (2, 'Lin');
insert into lesson_posts values (10, 1, 'Hello'), (11, 1, 'Launch');
select count(*) from lesson_posts;
```

PRO TIP Postlight references auth.users with UUID identifiers.

Run and explain example 1

EXPECTED RESULT

count = 2

WHY IT BEHAVES THIS WAY

author_id references an existing author. The foreign key rejects an unknown author.

Worked example 2: Join posts to authors

workbook.sql

```sql
select p.title, a.name as author
from lesson_posts p
join lesson_authors a on a.id = p.author_id
order by p.id;
```

PRO TIP An inner join omits rows with no matching partner.

Run and explain example 2

EXPECTED RESULT

Hello / Ada
Launch / Ada

WHY IT BEHAVES THIS WAY

The ON condition matches each post to its author. Aliases shorten qualified column names.

Worked example 3: Rollback part of a transaction

workbook.sql

```
savepoint before_edit;
update lesson_posts set title = 'Temporary' where id = 10;
rollback to savepoint before_edit;
select title from lesson_posts where id = 10;
```

PRO TIP In Python, a successful connection context commits; an exception rolls back.

Run and explain example 3

EXPECTED RESULT

title = Hello

WHY IT BEHAVES THIS WAY

Rolling back to the savepoint undoes the update while keeping earlier work.

Worked example 4: Index and ownership predicate

workbook.sql

```
01    create index lesson_posts_author_idx on lesson_posts(author_id);
02    select id from lesson_posts
03    where author_id = 1 order by id;
```

PRO TIP A WHERE condition is not a replacement for authenticating the requester.

Run and explain example 4

EXPECTED RESULT

10
11

WHY IT BEHAVES THIS WAY

The predicate selects author 1 records. An index can help lookups as tables grow.

Exercise 1: Authors with no posts

YOUR TASK

Return every author and their post count, including zero.

CHECK THESE CASES

Ada has 2; Lin has 0.

HINT AFTER A TRY

Use LEFT JOIN and count(p.id), not count(*).

Exercise 2: Atomic edit

YOUR TASK

Change one title, inspect it, then roll back the change using a savepoint.

CHECK THESE CASES

The final title is still Hello.

HINT AFTER A TRY

Put a savepoint before UPDATE.

Checkpoint before moving on

01 Run and explain all four worked examples. Match their actual results to the expected behavior.

02 Attempt both exercises before reading the solutions. Verify every acceptance case.

03 Explain how this chapter supports Postlight: Understand posts, authors, post_chunks and safe database writes.

Official reference